THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

Jun 2, 2020 9:30 a.m. – 12:00 noon

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL SIX questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (24 points)

Prove or disprove each of the following statements:

(a) For any nonempty sets A, B, C, D,

$$(A \times B) \cup (C \times D) = (A \cup C) \times (B \cup D)$$

(b) For any subset A of the set $\mathbb{I}$ of irrational numbers, if A is infinite, then A is uncountable.

(c) For any nonempty sets A, B and for any function $f : A \longrightarrow B$, if $X = f^{-1}(f(X))$ for any subset X of A, then f is injective.

(d) For any nonempty sets A, B and for any function $f : A \longrightarrow B$, if $f(f^{-1}(Y)) = Y$ for any subset Y of B, then f is surjective.

2. (10 points) Consider the sequence $a_1, a_2, a_3, \ldots$ of numbers satisfying $a_1 = 4$, $a_2 = 14$ and

$$a_n = 4a_{n-1} - a_{n-2} \quad \text{for any integer } n \geq 3.$$

Find a general formula for a_n and prove by induction that your answer is correct.

(*Hint*: Consider the quadratic equation $x^2 - 4x + 1 = 0$.)

3. (14 points) Let R be the relation on $\mathbb{R}^2$ defined by

$$(a, b)R(c, d) \text{ if } (ab = 0 \text{ and } cd = 0) \text{ or } (ac > 0 \text{ and } bd > 0).$$

(a) Prove that R is an equivalence relation.

(b) Find all the distinct equivalence classes and hence give a partition of $\mathbb{R}^2$ with respect to R.

4. (16 points) Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ be a function satisfying

$$f(yf(x) + y) \;=\; xy + f(y) \quad \text{for any } x, y \in \mathbb{R}. \qquad (*)$$

(a) Prove that f is injective. Hence show that $f(0) = 0$.

(b) Prove that f is surjective.

(c) Hence, or otherwise, find all functions $f : \mathbb{R} \longrightarrow \mathbb{R}$ satisfying $(*)$.

5. (20 points)

(a) Prove that for any positive integer m,

$$5^m \equiv 2^m \equiv \begin{cases} 1 \pmod 3 & \text{if } m \text{ is even,} \\ 2 \pmod 3 & \text{if } m \text{ is odd.} \end{cases}$$

(b) Prove that if there exist positive integers x, y, z satisfying the equation

$$3^x + 4^y = 5^z,$$

then z must be even. Write $z = 2k$ for some positive integer k. Prove that k must be odd and y must be even.

(c) Hence, or otherwise, prove that $(x, y, z) = (2, 2, 2)$ is the only positive integral solution of the equation $3^x + 4^y = 5^z$.
(*Hint*: Use (b) and consider modulo 8.)

6. (16 points)

(a) Use the Schröder-Bernstein Theorem to prove that $\mathbb{N}^m$ is denumerable for any $m \in \mathbb{N}$.
(*Hint*: For the function from $\mathbb{N}^m$ to $\mathbb{N}$, use the Fundamental Theorem of Arithmetic to help you to construct an injective function.)

(b) For any $m \in \mathbb{N}$, a function $f : \mathbb{Z} \longrightarrow \mathbb{N}$ is said to be *periodic with period* m if $f(x+m) = f(x)$ for any $x \in \mathbb{Z}$. Denote by P_m the set of all periodic functions $f : \mathbb{Z} \longrightarrow \mathbb{N}$ with period m. Use (a) and the Schröder-Bernstein Theorem to prove that P_m is denumerable for any $m \in \mathbb{N}$.

(*Hint*: First show that $|P_m| = |\mathbb{N}^m|$. Construct an injective function $\Phi : P_m \longrightarrow \mathbb{N}^m$ and an injective function $\Psi : \mathbb{N}^m \longrightarrow P_m$.)

******* **END OF PAPER** *******